Online Train Reservation System

Aim

To design and develop an online train reservation system that lets passengers search trains, book tickets, check PNR status and cancel tickets with refund, and lets the admin maintain train, fare and payment details.

Algorithm

1. Start the application; the passenger logs in with valid credentials (registers first if new).
2. Passenger enters the journey date, origin station, destination station, class type and passenger details.
3. The system displays the train list with seat availability and via details; the passenger selects a train.
4. Passenger pays online or offline; after successful payment the system generates the ticket and PNR number and sends the payment details to the admin.
5. Passenger checks the PNR status (confirmed, RAC, waiting list) and prints the ticket from the dashboard.
6. To cancel, the passenger enters the PNR number; refund = fare × refundable percentage based on the reservation and cancellation dates, approved by the admin.
7. Admin logs in and stores and updates train details, schedules and fares by zone, class and date.
8. Admin manages users and payments and logs out.

CODE:

HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>RailConnect | Train Reservation</title>
  <link rel="stylesheet" href="Train.css">
</head>
<body>

<header class="navbar">
  <a class="logo" href="#home">🚆 RailConnect</a>
  <nav>
    <a href="#home">Home</a>
```

```
      <a href="#search">Search Trains</a>
      <a href="#book">Book Ticket</a>
      <a href="#pnr">PNR Status</a>
      <a href="#mytickets">My Tickets</a>
      <a href="#account">Account</a>
      <a href="#admin">Admin</a>
    </nav>
</header>

<section class="hero" id="home">
    <div class="hero-content">
      <span class="tag">YOUR JOURNEY, OUR PRIORITY</span>
      <h1>Travel More.<br><span>Worry Less.</span></h1>
      <p>Search trains, check seat availability and reserve your next journey with ease.</p>
      <a class="primary-link" href="#search">Book Your Journey →</a>
    </div>
    <div class="hero-art">
      <div class="train-emoji">🚆</div>
      <div class="track"></div>
      <p>Comfortable journeys start here.</p>
    </div>
</section>

<section class="stats">
    <div class="stat-card">
      <span>🚆</span>
      <div><h3 id="trainCount">4</h3><p>Demo Trains</p></div>
    </div>
    <div class="stat-card">
      <span>🎫</span>
      <div><h3>Easy</h3><p>Ticket Booking</p></div>
    </div>
    <div class="stat-card">
      <span>🔎</span>
      <div><h3>Instant</h3><p>PNR Checking</p></div>
    </div>
```

```
</section>

<section id="search">
  <div class="section-heading">
    <span class="tag">PLAN YOUR TRIP</span>
    <h2>Find Your Train</h2>
    <p>Enter your journey details to explore available trains.</p>
  </div>

  <form id="searchForm" class="search-form">
    <div>
      <label for="fromStation">From Station</label>
      <input id="fromStation" placeholder="e.g. Chennai" required>
    </div>
    <div>
      <label for="toStation">To Station</label>
      <input id="toStation" placeholder="e.g. Bengaluru" required>
    </div>
    <div>
      <label for="journeyDate">Journey Date</label>
      <input type="date" id="journeyDate" required>
    </div>
    <button type="submit">Search Trains</button>
  </form>

  <div id="searchMessage" class="message"></div>
  <div id="trainResults" class="train-grid"></div>
</section>

<section id="book" class="soft-section">
  <div class="section-heading">
    <span class="tag">RESERVE YOUR SEAT</span>
    <h2>Book Your Ticket</h2>
    <p>Log in first, select a train and enter passenger details.</p>
  </div>
```

```
<form id="bookingForm" class="form-card">
  <div class="form-grid">
    <div>
      <label for="passengerName">Passenger Name</label>
      <input id="passengerName" required>
    </div>
    <div>
      <label for="passengerAge">Age</label>
      <input id="passengerAge" type="number" min="1" max="120" required>
    </div>
    <div>
      <label for="passengerGender">Gender</label>
      <select id="passengerGender" required>
        <option value="">Select gender</option>
        <option>Female</option>
        <option>Male</option>
        <option>Other</option>
      </select>
    </div>
    <div>
      <label for="passengerEmail">Email</label>
      <input id="passengerEmail" type="email" required>
    </div>
    <div>
      <label for="selectedTrain">Train</label>
      <select id="selectedTrain" required>
        <option value="">Select a train</option>
      </select>
    </div>
    <div>
      <label for="bookingDate">Journey Date</label>
      <input type="date" id="bookingDate" required>
    </div>
    <div>
      <label for="travelClass">Travel Class</label>
      <select id="travelClass" required>
```

```
                <option value="Sleeper">Sleeper</option>
                <option value="AC 3 Tier">AC 3 Tier</option>
                <option value="AC 2 Tier">AC 2 Tier</option>
                <option value="First Class">First Class</option>
              </select>
            </div>
            <div>
              <label for="paymentMode">Payment Mode</label>
              <select id="paymentMode" required>
                <option value="Online Demo">Online Payment (Demo)</option>
                <option value="Offline">Offline Payment (Demo)</option>
              </select>
            </div>
          </div>

          <div class="notice">
            <strong>Fare information</strong>
            <p id="fareInfo">Choose a train to view the fare.</p>
          </div>
          <button type="submit">Confirm Reservation</button>
          <p id="bookingMessage" class="message"></p>
        </form>
      </section>

      <section id="pnr">
        <div class="section-heading">
          <span class="tag">CHECK YOUR RESERVATION</span>
          <h2>PNR Status</h2>
          <p>Enter your PNR number to check your ticket status.</p>
        </div>

        <form id="pnrForm" class="compact-form">
          <label for="pnrInput">PNR Number</label>
          <input id="pnrInput" placeholder="Enter your PNR" required>
          <button type="submit">Check Status</button>
        </form>
```

```
    <div id="pnrResult"></div>
</section>

<section id="mytickets" class="soft-section">
    <div class="section-heading">
        <span class="tag">YOUR TRAVEL RECORDS</span>
        <h2>My Tickets</h2>
        <p>View, print or cancel your reservations.</p>
    </div>

    <div id="ticketList"></div>

    <div class="cancel-panel">
        <h3>Cancel Ticket Using PNR</h3>
        <form id="cancelForm">
            <label for="cancelPnr">PNR Number</label>
            <input id="cancelPnr" required placeholder="Enter PNR number">
            <button type="submit" class="danger-button">Cancel Ticket</button>
        </form>
        <p id="cancelMessage" class="message"></p>
    </div>
</section>

<section id="account">
    <div class="section-heading">
        <span class="tag">PASSENGER ACCOUNT</span>
        <h2>Register or Login</h2>
        <p>Create an account to reserve and manage tickets.</p>
    </div>

    <div class="account-grid">
        <form id="registerForm" class="form-card">
            <h3>Create Account</h3>
            <label for="regName">Full Name</label>
            <input id="regName" required>
            <label for="regEmail">Email</label>
```

```
        <input id="regEmail" type="email" required>
        <label for="regPassword">Password</label>
        <input id="regPassword" type="password" minlength="4" required>
        <button type="submit">Register</button>
      </form>

      <form id="loginForm" class="form-card">
        <h3>Passenger Login</h3>
        <label for="loginEmail">Email</label>
        <input id="loginEmail" type="email" required>
        <label for="loginPassword">Password</label>
        <input id="loginPassword" type="password" required>
        <button type="submit">Login</button>
        <button type="button" class="light-button" onclick="forgotPassword()">Forgot
Password?</button>
      </form>
    </div>
    <p id="accountMessage" class="message"></p>
    <button type="button" class="light-button" onclick="logout()">Logout</button>
</section>

<section id="admin" class="soft-section">
    <div class="section-heading">
      <span class="tag">SYSTEM MANAGEMENT</span>
      <h2>Administrator Panel</h2>
      <p>Manage trains, fares, seat availability and reservation reports.</p>
    </div>

    <div class="admin-login">
      <p><strong>Demo Admin:</strong> admin@railconnect.com</p>
      <p><strong>Password:</strong> rail123</p>
      <form id="adminLoginForm">
        <label for="adminEmail">Admin Email</label>
        <input id="adminEmail" type="email" required>
        <label for="adminPassword">Password</label>
        <input id="adminPassword" type="password" required>
```

```
      <button type="submit">Admin Login</button>
    </form>
    <p id="adminMessage" class="message"></p>
  </div>

  <div id="adminPanel" hidden>
    <div class="admin-box">
      <h3>Add a Train</h3>
      <form id="trainForm" class="form-card">
        <label for="trainName">Train Name</label>
        <input id="trainName" required>
        <label for="trainNumber">Train Number</label>
        <input id="trainNumber" required>
        <label for="adminFrom">Origin Station</label>
        <input id="adminFrom" required>
        <label for="adminTo">Destination Station</label>
        <input id="adminTo" required>
        <label for="trainTime">Departure Time</label>
        <input id="trainTime" type="time" required>
        <label for="trainSeats">Total Seats</label>
        <input id="trainSeats" type="number" min="1" required>
        <label for="trainFare">Base Fare (₹)</label>
        <input id="trainFare" type="number" min="1" required>
        <button type="submit">Add Train</button>
      </form>
    </div>

    <div class="admin-box">
      <h3>Manage Trains</h3>
      <div id="adminTrainList"></div>
    </div>

    <div class="admin-box">
      <h3>All Reservations</h3>
      <div id="adminBookingList"></div>
    </div>
```

```
      <div class="admin-box">
        <h3>Reservation Reports</h3>
        <button type="button" onclick="generateReport()">Generate Report</button>
        <button type="button" class="light-button" onclick="downloadReport()">Download
CSV</button>
        <div id="reportResult"></div>
      </div>
    </div>
</section>

<footer>
    <h3>🚆 RailConnect</h3>
    <p>Your journey begins with a single reservation.</p>
    <p>College Lab Demonstration · 2026</p>
</footer>

<script src="script.js"></script>
</body>
</html>
```

CSS:

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: Arial, sans-serif;
    background: #f5f7fc;
    color: #26334d;
    line-height: 1.6;
}

.navbar {
    background: linear-gradient(100deg, #101d42, #263f78);
    padding: 18px 5%;
```

```
    display: flex;
    justify-content: space-between;
    align-items: center;
    gap: 15px;
    flex-wrap: wrap;
    position: sticky;
    top: 0;
    z-index: 10;
    box-shadow: 0 4px 18px #101d4220;
}

.logo {
    color: white;
    font-size: 25px;
    font-weight: bold;
    text-decoration: none;
}

nav {
    display: flex;
    flex-wrap: wrap;
    gap: 5px;
}

nav a {
    color: white;
    text-decoration: none;
    padding: 8px 11px;
    border-radius: 8px;
    font-size: 14px;
}

nav a:hover {
    background: #f28b35;
}

.hero {
    padding: 65px 8%;
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 30px;
    flex-wrap: wrap;
    background: linear-gradient(120deg, #e0eaff, #fcebdc, #f4e5ff);
}

.hero-content {
```

```
    flex: 1;
    min-width: 260px;
}

.tag {
    display: inline-block;
    color: #d96820;
    background: #fff0e3;
    padding: 6px 12px;
    border-radius: 20px;
    font-size: 12px;
    font-weight: bold;
    letter-spacing: 1px;
}

.hero h1 {
    font-size: clamp(38px, 6vw, 62px);
    line-height: 1.15;
    margin: 18px 0;
    color: #16284e;
}

.hero h1 span {
    color: #e7792b;
}

.hero p {
    max-width: 500px;
    margin-bottom: 25px;
    color: #56627b;
    font-size: 17px;
}

.primary-link {
    display: inline-block;
    background: #e7792b;
    color: white;
    text-decoration: none;
    padding: 13px 22px;
    border-radius: 10px;
    font-weight: bold;
}

.primary-link:hover {
    background: #c65a14;
}

.hero-art {
```

```
    flex: 1;
    min-width: 220px;
    text-align: center;
    background: #ffffffb8;
    padding: 35px 20px;
    border-radius: 25px;
    box-shadow: 0 12px 35px #263f7815;
}

.train-emoji {
    font-size: 100px;
    margin-bottom: 10px;
}

.track {
    width: 80%;
    height: 5px;
    background: repeating-linear-gradient(
        90deg, #e7792b 0 20px, transparent 20px 28px
    );
    margin: 0 auto 18px;
}

.stats {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(180px, 1fr));
    gap: 18px;
    padding: 25px 6%;
}

.stat-card {
    display: flex;
    align-items: center;
    gap: 15px;
    background: white;
    border-radius: 16px;
    padding: 20px;
    box-shadow: 0 5px 18px #263f780d;
}

.stat-card > span {
    background: #fff0e3;
    padding: 12px;
    border-radius: 12px;
    font-size: 25px;
}
```

```
.stat-card h3 {
    color: #253e78;
    font-size: 23px;
}

.stat-card p {
    color: #7b8496;
    font-size: 13px;
}

section:not(.hero):not(.stats) {
    padding: 45px 6%;
    scroll-margin-top: 80px;
}

.section-heading {
    text-align: center;
    margin-bottom: 25px;
}

.section-heading h2 {
    color: #1d315d;
    font-size: 30px;
    margin: 10px 0;
}

.section-heading p {
    color: #758096;
}

.soft-section {
    background: #eaf0fc;
}

form {
    background: white;
    padding: 23px;
    border-radius: 17px;
    box-shadow: 0 6px 22px #263f7810;
}

.search-form {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(180px, 1fr));
    gap: 17px;
    align-items: end;
}
```

```
label {
    display: block;
    color: #34476d;
    font-size: 14px;
    font-weight: bold;
    margin: 10px 0 6px;
}

input, select, textarea {
    width: 100%;
    padding: 12px;
    border: 1px solid #d9e0ee;
    border-radius: 9px;
    background: #fbfcff;
    color: #26334d;
    font: inherit;
    font-size: 14px;
}

input:focus, select:focus, textarea:focus {
    outline: 2px solid #b9cafa;
    border-color: #607fcb;
}

button {
    border: none;
    background: linear-gradient(100deg, #e7792b, #f49a4c);
    color: white;
    padding: 12px 18px;
    margin-top: 12px;
    border-radius: 9px;
    font-size: 14px;
    font-weight: bold;
    cursor: pointer;
    transition: 0.2s;
}

button:hover {
    transform: translateY(-2px);
    filter: brightness(0.94);
}

.search-form button {
    width: 100%;
}

.train-grid {
```

```
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
    gap: 20px;
    margin-top: 25px;
}

.train-card {
    background: white;
    padding: 22px;
    border-radius: 17px;
    border-top: 5px solid #e7792b;
    box-shadow: 0 6px 20px #263f7812;
    transition: 0.25s;
}

.train-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 12px 28px #263f7820;
}

.train-card h3 {
    color: #243d75;
    margin: 8px 0;
}

.train-number {
    display: inline-block;
    background: #eaf0ff;
    color: #3158a6;
    padding: 5px 9px;
    border-radius: 7px;
    font-size: 12px;
    font-weight: bold;
}

.route {
    font-size: 17px;
    font-weight: bold;
    margin: 12px 0;
}

.fare {
    color: #df7026;
    font-size: 22px;
    font-weight: bold;
}

.form-card {
```

```
    max-width: 800px;
    margin: 0 auto;
}

.form-card h3 {
    color: #243d75;
    margin-bottom: 15px;
}

.form-grid {
    display: grid;
    grid-template-columns: repeat(2, minmax(0, 1fr));
    gap: 12px 18px;
}

.notice {
    background: #fff4e8;
    border-left: 4px solid #e7792b;
    padding: 14px;
    border-radius: 9px;
    margin-top: 20px;
}

.message {
    margin-top: 14px;
    color: #285d43;
    font-weight: bold;
    overflow-wrap: anywhere;
}

.compact-form {
    max-width: 500px;
    margin: 0 auto;
}

#pnrResult {
    max-width: 600px;
    margin: 20px auto;
}

.ticket-card {
    background: white;
    border-radius: 16px;
    padding: 22px;
    margin: 15px auto;
    max-width: 850px;
    border-left: 5px solid #e7792b;
    box-shadow: 0 5px 20px #263f7810;
```

```
    overflow-wrap: anywhere;
}

.ticket-card h3 {
    color: #243d75;
    margin-bottom: 10px;
}

.status {
    display: inline-block;
    padding: 5px 10px;
    border-radius: 20px;
    background: #dff7e8;
    color: #17663a;
    font-size: 12px;
    font-weight: bold;
}

.cancelled {
    background: #ffe1e1;
    color: #a72b2b;
}

.cancel-panel {
    max-width: 600px;
    margin: 35px auto 0;
}

.cancel-panel h3 {
    color: #243d75;
}

.danger-button {
    background: #d94848;
}

.account-grid {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 22px;
    max-width: 950px;
    margin: 0 auto;
}

.account-grid .form-card {
    width: 100%;
}
```

```
.light-button {
   background: #e6ecfb;
   color: #284477;
}

.admin-login {
   max-width: 550px;
   margin: 0 auto;
}

.admin-login > p {
   text-align: center;
   margin-bottom: 8px;
}

#adminPanel[hidden] {
   display: none;
}

.admin-box {
   background: white;
   border-radius: 17px;
   padding: 22px;
   margin-top: 25px;
   box-shadow: 0 5px 20px #263f7810;
   overflow-wrap: anywhere;
}

.admin-box h3 {
   color: #243d75;
   margin-bottom: 15px;
}

.admin-item {
   border: 1px solid #e0e6f1;
   border-radius: 10px;
   padding: 15px;
   margin: 12px 0;
}

footer {
   background: #101d42;
   color: white;
   text-align: center;
   padding: 30px 15px;
}

footer h3 {
```

```
    margin-bottom: 8px;
}

@media (max-width: 700px) {
    .navbar {
        position: static;
        flex-direction: column;
        align-items: flex-start;
    }

    nav {
        gap: 5px;
    }

    nav a {
        font-size: 12px;
        padding: 7px;
    }

    .hero {
        padding: 40px 6%;
    }

    .hero-art {
        width: 100%;
    }

    .form-grid {
        grid-template-columns: 1fr;
    }

    section:not(.hero):not(.stats) {
        padding: 30px 5%;
    }

    .section-heading h2 {
        font-size: 25px;
    }
}

@media print {
    body * {
        visibility: hidden;
    }

    .ticket-card, .ticket-card * {
        visibility: visible;
    }
```

```
    .ticket-card {
        position: absolute;
        left: 0;
        top: 0;
        width: 100%;
        box-shadow: none;
        border: 1px solid #333;
    }

    button, header, footer {
        display: none !important;
    }
}
```

JAVASCRIPT:

```
const defaultTrains = [
    { id: 1, number: "12601", name: "Chennai Express", from: "Chennai", to: "Bengaluru", time:
"06:30", seats: 40, fare: 450 },
    { id: 2, number: "12603", name: "Coastal Express", from: "Chennai", to: "Hyderabad", time:
"09:15", seats: 35, fare: 600 },
    { id: 3, number: "12605", name: "Southern Express", from: "Chennai", to: "Madurai", time:
"21:00", seats: 30, fare: 350 },
    { id: 4, number: "12607", name: "City Intercity", from: "Bengaluru", to: "Chennai", time:
"14:00", seats: 45, fare: 400 }
];

function load(key, fallback) {
    try {
        const data = localStorage.getItem(key);
        return data ? JSON.parse(data) : fallback;
    } catch {
        return fallback;
    }
}

let trains = load("railTrains", defaultTrains);
let users = load("railUsers", []);
let reservations = load("railReservations", []);
let currentUser = load("railCurrentUser", null);
let adminLoggedIn = false;

function save() {
    localStorage.setItem("railTrains", JSON.stringify(trains));
    localStorage.setItem("railUsers", JSON.stringify(users));
    localStorage.setItem("railReservations", JSON.stringify(reservations));
    localStorage.setItem("railCurrentUser", JSON.stringify(currentUser));
}
```

```
function today() {
   const d = new Date();
   return d.getFullYear() + "-" +
      String(d.getMonth() + 1).padStart(2, "0") + "-" +
      String(d.getDate()).padStart(2, "0");
}

document.getElementById("journeyDate").min = today();
document.getElementById("bookingDate").min = today();

function escapeHTML(value) {
   return String(value ?? "").replace(/[&<>"']/g, c => ({
      "&": "&amp;",
      "<": "&lt;",
      ">": "&gt;",
      '"': "&quot;",
      "'": "&#39;"
   })[c]);
}

function findTrain(id) {
   return trains.find(t => t.id === Number(id));
}

function getFare(train, travelClass) {
   const multipliers = {
      "Sleeper": 1,
      "AC 3 Tier": 1.6,
      "AC 2 Tier": 2.2,
      "First Class": 3
   };

   return Math.round(train.fare * multipliers[travelClass]);
}

function seatsBooked(trainId, date) {
   return reservations.filter(r =>
      r.trainId === Number(trainId) &&
      r.date === date &&
      r.status !== "Cancelled" &&
      r.status !== "Refunded"
   ).length;
}

function getStatus(trainId, date) {
   const train = findTrain(trainId);
   if (!train) return "Waiting List";
```

```
    const used = seatsBooked(trainId, date);

    if (used < train.seats) return "Confirmed";
    if (used < train.seats + 10) return "RAC";
    return "Waiting List";
}

function updateTrainOptions() {
    const select = document.getElementById("selectedTrain");
    const oldValue = select.value;

    select.innerHTML = '<option value="">Select a train</option>';

    trains.forEach(train => {
        const option = document.createElement("option");
        option.value = train.id;
        option.textContent =
            train.number + " - " + train.name +
            " (" + train.from + " to " + train.to + ")";
        select.appendChild(option);
    });

    if ([...select.options].some(o => o.value === oldValue)) {
        select.value = oldValue;
    }

    document.getElementById("trainCount").textContent = trains.length;
}

function showTrains(list, date) {
    const area = document.getElementById("trainResults");
    area.innerHTML = "";

    if (!list.length) {
        area.textContent = "No trains found for this route.";
        return;
    }

    list.forEach(train => {
        const status = date ? getStatus(train.id, date) : "Check date";
        const card = document.createElement("div");
        card.className = "train-card";

        card.innerHTML = `
            <span class="train-number">TRAIN ${escapeHTML(train.number)}</span>
            <h3>${escapeHTML(train.name)}</h3>
            <p class="route">${escapeHTML(train.from)} → ${escapeHTML(train.to)}</p>
```

```
            <p>Departure: ${escapeHTML(train.time)}</p>
            <p>Total seats: ${train.seats}</p>
            <p>Availability: ${escapeHTML(status)}</p>
            <p class="fare">From ₹${train.fare}</p>
        `;

        const button = document.createElement("button");
        button.textContent = "Book This Train";
        button.onclick = () => {
            document.getElementById("selectedTrain").value = train.id;
            if (date) {
                document.getElementById("bookingDate").value = date;
            }
            updateFare();
            document.getElementById("book").scrollIntoView({ behavior: "smooth" });
        };

        card.appendChild(button);
        area.appendChild(card);
    });
}

document.getElementById("searchForm").addEventListener("submit", function(e) {
    e.preventDefault();

    const from = document.getElementById("fromStation").value.trim().toLowerCase();
    const to = document.getElementById("toStation").value.trim().toLowerCase();
    const date = document.getElementById("journeyDate").value;

    if (date < today()) {
        document.getElementById("searchMessage").textContent =
            "Please choose today or a future date.";
        return;
    }

    const results = trains.filter(t =>
        t.from.toLowerCase() === from &&
        t.to.toLowerCase() === to
    );

    document.getElementById("searchMessage").textContent =
        results.length + " train(s) found.";

    showTrains(results, date);
});

document.getElementById("selectedTrain").addEventListener("change", updateFare);
document.getElementById("travelClass").addEventListener("change", updateFare);
```

```
function updateFare() {
  const train = findTrain(document.getElementById("selectedTrain").value);
  const travelClass = document.getElementById("travelClass").value;

  document.getElementById("fareInfo").textContent = train
    ? "Estimated fare: ₹" + getFare(train, travelClass) +
      " per passenger. Final status depends on availability."
    : "Choose a train to view the fare.";
}

// USER REGISTRATION
document.getElementById("registerForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const name = document.getElementById("regName").value.trim();
  const email = document.getElementById("regEmail").value.trim().toLowerCase();
  const password = document.getElementById("regPassword").value;

  if (users.some(u => u.email === email)) {
    document.getElementById("accountMessage").textContent =
      "This email is already registered.";
    return;
  }

  users.push({ name, email, password });
  currentUser = { name, email };
  save();
  renderTickets();

  document.getElementById("accountMessage").textContent =
    "Registration successful. You are logged in.";
  this.reset();
});

// USER LOGIN
document.getElementById("loginForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const email = document.getElementById("loginEmail").value.trim().toLowerCase();
  const password = document.getElementById("loginPassword").value;

  const user = users.find(u =>
    u.email === email && u.password === password
  );

  if (!user) {
    document.getElementById("accountMessage").textContent =
```

```
            "Invalid credentials. Please register or try again.";
        return;
    }

    currentUser = { name: user.name, email: user.email };
    save();
    renderTickets();

    document.getElementById("accountMessage").textContent =
        "Welcome, " + user.name + "!";
    this.reset();
});

function logout() {
    currentUser = null;
    save();
    renderTickets();
    document.getElementById("accountMessage").textContent = "You have logged out.";
}

function forgotPassword() {
    const email = prompt("Enter your registered email:");
    if (!email) return;

    const user = users.find(u =>
        u.email === email.trim().toLowerCase()
    );

    alert(user
        ? "Demo only: password reset requires a secure email service."
        : "No account found for this email.");
}

// BOOK A TICKET
document.getElementById("bookingForm").addEventListener("submit", function(e) {
    e.preventDefault();

    if (!currentUser) {
        document.getElementById("bookingMessage").textContent =
            "Please log in before booking a ticket.";
        return;
    }

    const name = document.getElementById("passengerName").value.trim();
    const age = Number(document.getElementById("passengerAge").value);
    const gender = document.getElementById("passengerGender").value;
    const email = document.getElementById("passengerEmail").value.trim().toLowerCase();
    const train = findTrain(document.getElementById("selectedTrain").value);
```

```
const date = document.getElementById("bookingDate").value;
const travelClass = document.getElementById("travelClass").value;
const paymentMode = document.getElementById("paymentMode").value;

if (!train || !date || date < today() || age < 1 || age > 120) {
    document.getElementById("bookingMessage").textContent =
        "Please enter valid passenger, train and journey details.";
    return;
}

if (email !== currentUser.email) {
    document.getElementById("bookingMessage").textContent =
        "Please enter the email address of your logged-in account.";
    return;
}

const status = getStatus(train.id, date);
const fare = getFare(train, travelClass);

// Demo payment: confirmation is simulated.
if (!confirm("Demo payment of ₹" + fare +
    " using " + paymentMode + ". Continue with booking?")) {
    return;
}

const pnr = String(Date.now()).slice(-10);

const ticket = {
    pnr,
    userEmail: currentUser.email,
    name,
    age,
    gender,
    email,
    trainId: train.id,
    trainNumber: train.number,
    trainName: train.name,
    from: train.from,
    to: train.to,
    time: train.time,
    date,
    travelClass,
    fare,
    paymentMode,
    paymentStatus: "Demo Payment Recorded",
    status,
    refund: 0
```

```
    };

    reservations.push(ticket);
    save();
    renderTickets();
    renderAdmin();

    document.getElementById("bookingMessage").textContent =
      "Reservation created! PNR: " + pnr +
      " | Status: " + status +
      " | Fare: ₹" + fare +
      ". Demo payment only; no money was transferred.";

    this.reset();
    updateFare();
});

// TICKET HISTORY AND PRINTING
function renderTickets() {
    const area = document.getElementById("ticketList");
    area.innerHTML = "";

    if (!currentUser) {
      area.textContent = "Please log in to view your tickets.";
      return;
    }

    const mine = reservations.filter(r => r.userEmail === currentUser.email);

    if (!mine.length) {
      area.textContent = "No reservations found.";
      return;
    }

    mine.forEach(ticket => {
      const card = document.createElement("div");
      card.className = "ticket-card";

      card.innerHTML = `
        <h3>🚆 ${escapeHTML(ticket.trainName)}</h3>
        <p><b>PNR:</b> ${escapeHTML(ticket.pnr)}</p>
        <p><b>Passenger:</b> ${escapeHTML(ticket.name)} | Age: ${ticket.age}</p>
        <p><b>Route:</b> ${escapeHTML(ticket.from)} → ${escapeHTML(ticket.to)}</p>
        <p><b>Journey:</b> ${escapeHTML(ticket.date)} at ${escapeHTML(ticket.time)}</p>
        <p><b>Class:</b> ${escapeHTML(ticket.travelClass)}</p>
        <p><b>Fare:</b> ₹${ticket.fare}</p>
        <p><b>Payment:</b> ${escapeHTML(ticket.paymentStatus)}</p>
        <p><b>Reservation:</b>
```

```
            <span class="status ${ticket.status === "Cancelled" ? "cancelled" : ""}">
            ${escapeHTML(ticket.status)}</span>
          </p>
          ${ticket.refund ? "<p>Refund amount: ₹" + ticket.refund + "</p>" : ""}
        `;

        if (ticket.status !== "Cancelled" && ticket.status !== "Refunded") {
          const cancel = document.createElement("button");
          cancel.textContent = "Cancel Ticket";
          cancel.onclick = () => cancelReservation(ticket.pnr);
          card.appendChild(cancel);
        }

        const print = document.createElement("button");
        print.textContent = "Print Ticket";
        print.className = "light-button";
        print.onclick = () => window.print();
        card.appendChild(print);

        area.appendChild(card);
    });
}

// PNR STATUS CHECK
document.getElementById("pnrForm").addEventListener("submit", function(e) {
    e.preventDefault();

    const pnr = document.getElementById("pnrInput").value.trim();
    const ticket = reservations.find(r => r.pnr === pnr);
    const area = document.getElementById("pnrResult");

    if (!ticket) {
        area.textContent = "PNR not found. Please check the number.";
        return;
    }

    area.innerHTML = `
        <div class="ticket-card">
            <h3>PNR Status</h3>
            <p><b>PNR:</b> ${escapeHTML(ticket.pnr)}</p>
            <p><b>Train:</b> ${escapeHTML(ticket.trainName)}</p>
            <p><b>Passenger:</b> ${escapeHTML(ticket.name)}</p>
            <p><b>Route:</b> ${escapeHTML(ticket.from)} → ${escapeHTML(ticket.to)}</p>
            <p><b>Journey Date:</b> ${escapeHTML(ticket.date)}</p>
            <p><b>Status:</b> ${escapeHTML(ticket.status)}</p>
        </div>
    `;
});
```

```
// CANCELLATION AND DEMO REFUND RULES
function cancelReservation(pnr) {
  const ticket = reservations.find(r => r.pnr === pnr);

  if (!ticket || ticket.status === "Cancelled" || ticket.status === "Refunded") {
    alert("Ticket not found or already cancelled.");
    return;
  }

  if (!confirm("Cancel ticket PNR " + pnr + "?")) return;

  const todayDate = new Date(today() + "T00:00:00");
  const journey = new Date(ticket.date + "T00:00:00");
  const daysLeft = Math.floor((journey - todayDate) / 86400000);

  let percentage = 0;
  if (daysLeft >= 7) percentage = 90;
  else if (daysLeft >= 3) percentage = 75;
  else if (daysLeft >= 1) percentage = 50;

  ticket.refund = Math.round(ticket.fare * percentage / 100);
  ticket.status = "Cancelled";

  save();
  renderTickets();
  renderAdmin();

  document.getElementById("cancelMessage").textContent =
    "Ticket cancelled. Demo refundable amount: ₹" + ticket.refund +
    " (" + percentage + "%). No real payment or refund was processed.";
}

document.getElementById("cancelForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const pnr = document.getElementById("cancelPnr").value.trim();
  const ticket = reservations.find(r => r.pnr === pnr);

  if (!ticket) {
    document.getElementById("cancelMessage").textContent = "PNR not found.";
    return;
  }

  if (currentUser && ticket.userEmail !== currentUser.email && !adminLoggedIn) {
    document.getElementById("cancelMessage").textContent =
      "Log in with the account used to make this reservation.";
    return;
```

```
    }

    cancelReservation(pnr);
    this.reset();
});

// ADMIN LOGIN
document.getElementById("adminLoginForm").addEventListener("submit", function(e) {
    e.preventDefault();

    const email = document.getElementById("adminEmail").value.trim();
    const password = document.getElementById("adminPassword").value;

    if (email === "admin@railconnect.com" && password === "rail123") {
        adminLoggedIn = true;
        document.getElementById("adminPanel").hidden = false;
        document.getElementById("adminMessage").textContent =
            "Admin login successful.";
        renderAdmin();
    } else {
        document.getElementById("adminMessage").textContent =
            "Invalid admin credentials.";
    }
});

// ADMIN ADDS TRAIN
document.getElementById("trainForm").addEventListener("submit", function(e) {
    e.preventDefault();

    if (!adminLoggedIn) return;

    const number = document.getElementById("trainNumber").value.trim();

    if (trains.some(t => t.number === number)) {
        alert("This train number already exists.");
        return;
    }

    trains.push({
        id: Date.now(),
        number,
        name: document.getElementById("trainName").value.trim(),
        from: document.getElementById("adminFrom").value.trim(),
        to: document.getElementById("adminTo").value.trim(),
        time: document.getElementById("trainTime").value,
        seats: Number(document.getElementById("trainSeats").value),
        fare: Number(document.getElementById("trainFare").value)
    });
```

```
    save();
    updateTrainOptions();
    renderAdmin();
    this.reset();
    alert("Train added successfully.");
});

// ADMIN MANAGEMENT
function renderAdmin() {
    if (!adminLoggedIn) return;

    const trainArea = document.getElementById("adminTrainList");
    trainArea.innerHTML = "";

    trains.forEach(train => {
        const item = document.createElement("div");
        item.className = "admin-item";

        item.innerHTML = `
            <h4>${escapeHTML(train.number)} - ${escapeHTML(train.name)}</h4>
            <p>${escapeHTML(train.from)} → ${escapeHTML(train.to)}</p>
            <p>Departure: ${escapeHTML(train.time)}</p>
            <p>Seats: ${train.seats} | Base fare: ₹${train.fare}</p>
        `;

        const edit = document.createElement("button");
        edit.textContent = "Update Fare / Seats";
        edit.onclick = () => {
            const fare = prompt("Enter new base fare:", train.fare);
            if (fare === null) return;

            const seats = prompt("Enter new seat capacity:", train.seats);
            if (seats === null) return;

            const booked = reservations.filter(r =>
                r.trainId === train.id &&
                r.status !== "Cancelled" &&
                r.status !== "Refunded"
            ).length;

            if (Number(fare) <= 0 ||
                !Number.isInteger(Number(seats)) ||
                Number(seats) < 1 ||
                Number(seats) < booked) {
                alert("Invalid fare or seat capacity. Capacity cannot be less than existing
reservations.");
                return;
```

```
        }

        train.fare = Number(fare);
        train.seats = Number(seats);
        save();
        renderAdmin();
        updateTrainOptions();
    };

    const remove = document.createElement("button");
    remove.textContent = "Delete Train";
    remove.className = "danger-button";
    remove.onclick = () => {
        const active = reservations.some(r =>
            r.trainId === train.id &&
            r.status !== "Cancelled" &&
            r.status !== "Refunded"
        );

        if (active) {
            alert("This train has active reservations. Cancel them before deleting it.");
            return;
        }

        if (confirm("Delete this train?")) {
            trains = trains.filter(t => t.id !== train.id);
            save();
            updateTrainOptions();
            renderAdmin();
        }
    };

    item.appendChild(edit);
    item.appendChild(remove);
    trainArea.appendChild(item);
});

const bookingArea = document.getElementById("adminBookingList");
bookingArea.innerHTML = "";

reservations.forEach(ticket => {
    const item = document.createElement("div");
    item.className = "admin-item";
    item.innerHTML = `
        <p><b>PNR:</b> ${escapeHTML(ticket.pnr)}</p>
        <p>Passenger: ${escapeHTML(ticket.name)}</p>
        <p>Train: ${escapeHTML(ticket.trainName)}</p>
        <p>Date: ${escapeHTML(ticket.date)}</p>
```

```
            <p>Status: ${escapeHTML(ticket.status)}</p>
            <p>Payment: ${escapeHTML(ticket.paymentStatus)}</p>
        `;

        if (ticket.status !== "Cancelled" && ticket.status !== "Refunded") {
            const cancel = document.createElement("button");
            cancel.textContent = "Cancel Reservation";
            cancel.onclick = () => {
                if (confirm("Admin: cancel this reservation?")) {
                    ticket.status = "Cancelled";
                    ticket.refund = 0;
                    save();
                    renderAdmin();
                    renderTickets();
                }
            };
            item.appendChild(cancel);
        }

        bookingArea.appendChild(item);
    });
}

// ADMIN REPORT
function generateReport() {
    if (!adminLoggedIn) return;

    const active = reservations.filter(r =>
        r.status !== "Cancelled" && r.status !== "Refunded"
    );
    const cancelled = reservations.filter(r =>
        r.status === "Cancelled" || r.status === "Refunded"
    );
    const total = active.reduce((sum, r) => sum + r.fare, 0);

    document.getElementById("reportResult").innerHTML = `
        <div class="admin-item">
            <h3>Reservation Report</h3>
            <p>Total trains: ${trains.length}</p>
            <p>Registered users: ${users.length}</p>
            <p>Total reservations: ${reservations.length}</p>
            <p>Active reservations: ${active.length}</p>
            <p>Cancelled reservations: ${cancelled.length}</p>
            <p>Active ticket fare total: ₹${total}</p>
        </div>
    `;
}
```

```
function downloadReport() {
    if (!adminLoggedIn) {
        alert("Please log in as admin first.");
        return;
    }

    const rows = [
        ["PNR", "Passenger", "Email", "Train", "From", "To", "Date", "Class", "Fare", "Status"],
        ...reservations.map(r => [
            r.pnr, r.name, r.email, r.trainName,
            r.from, r.to, r.date, r.travelClass, r.fare, r.status
        ])
    ];

    const csv = rows.map(row =>
        row.map(value =>
            "" + String(value).replace(/"/g, """") + ""
        ).join(",")
    ).join("\r\n");

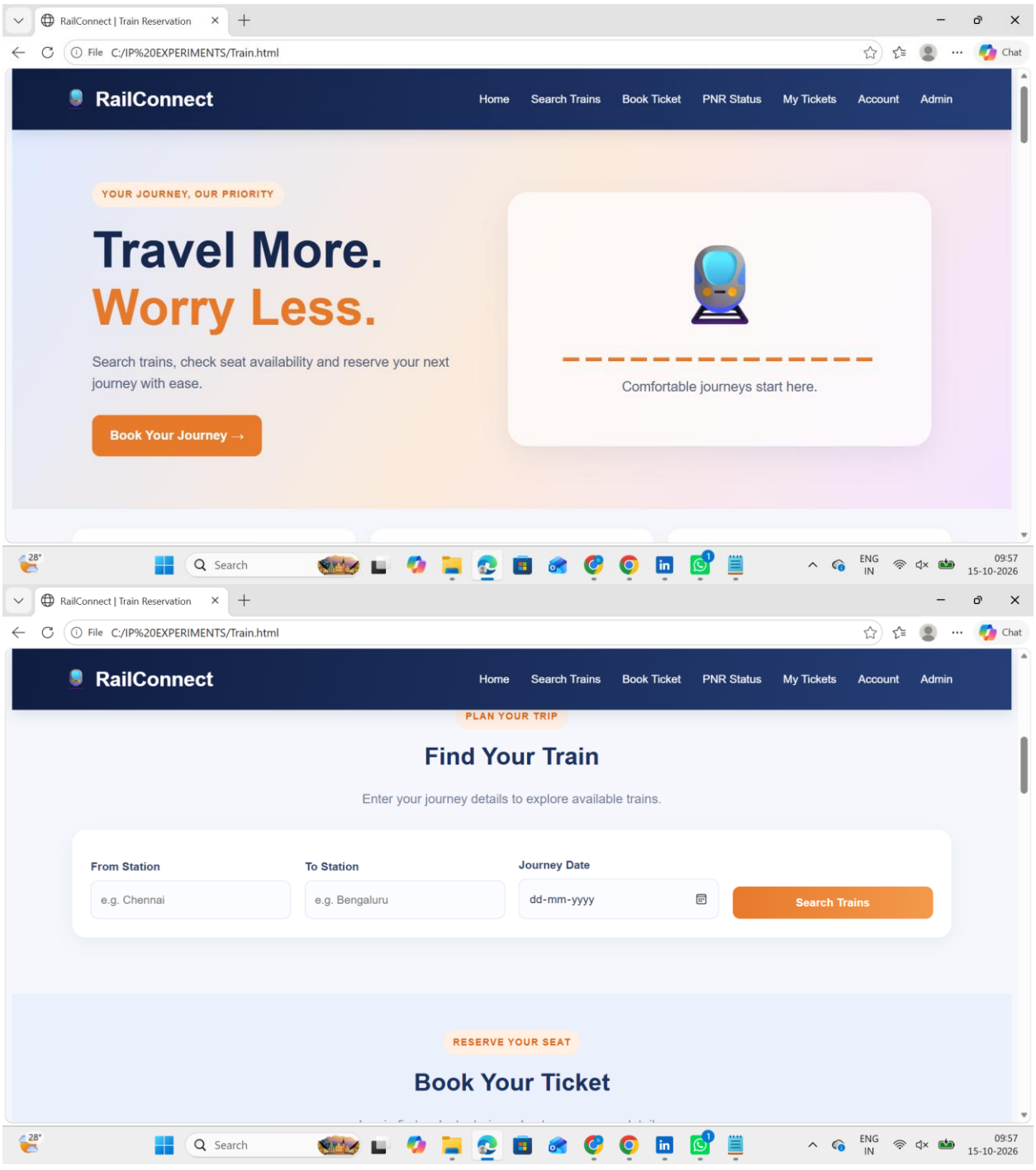
    const blob = new Blob([csv], { type: "text/csv;charset=utf-8;" });
    const url = URL.createObjectURL(blob);
    const link = document.createElement("a");
    link.href = url;
    link.download = "RailConnect_Report.csv";
    link.click();
    URL.revokeObjectURL(url);
}

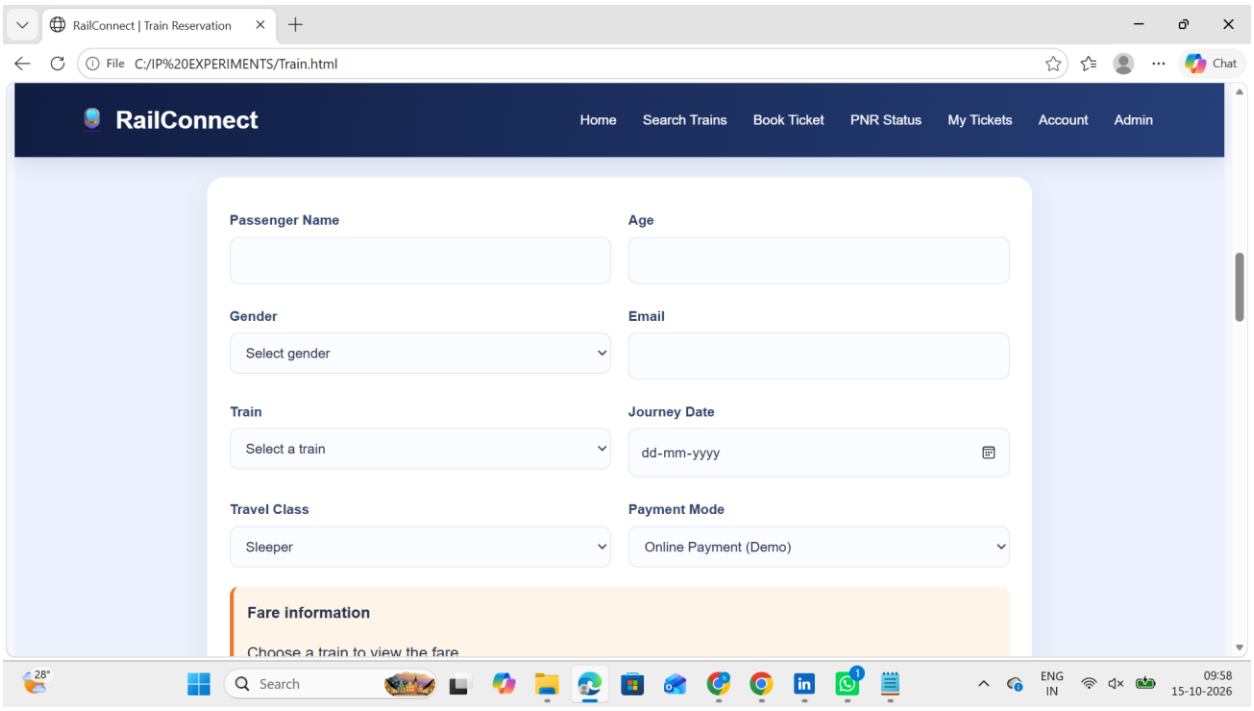
// INITIALIZE WEBSITE
updateTrainOptions();
showTrains(trains);
renderTickets();
```

Result

The train reservation system was developed and tested successfully. Passengers could search trains, book tickets, receive a PNR, check status, cancel and obtain refunds, and the admin could manage train, fare and payment details.

OUTPUT:


RailConnect | Train Reservation
File C:/IP%20EXPERIMENTS/Train.html
Chat
RailConnect
Home
Search Trains
Book Ticket
PNR Status
My Tickets
Account
Admin
YOUR JOURNEY, OUR PRIORITY
Travel More.
Worry Less.
Search trains, check seat availability and reserve your next journey with ease.
Book Your Journey →
Comfortable journeys start here.
Search
ENG
IN
09:57
15-10-2026
RailConnect | Train Reservation
File C:/IP%20EXPERIMENTS/Train.html
Chat
RailConnect
Home
Search Trains
Book Ticket
PNR Status
My Tickets
Account
Admin
PLAN YOUR TRIP
Find Your Train
Enter your journey details to explore available trains.
From Station
e.g. Chennai
To Station
e.g. Bengaluru
Journey Date
dd-mm-yyyy
Search Trains
RESERVE YOUR SEAT
Book Your Ticket
Search
ENG
IN
09:57
15-10-2026


RailConnect | Train Reservation
File C:/IP%20EXPERIMENTS/Train.html
Chat
RailConnect
Home
Search Trains
Book Ticket
PNR Status
My Tickets
Account
Admin
Passenger Name
Age
Gender
Select gender
Email
Train
Select a train
Journey Date
dd-mm-yyyy
Travel Class
Sleeper
Payment Mode
Online Payment (Demo)
Fare information
Choose a train to view the fare
28°
Search
ENG
IN
09:58
15-10-2026